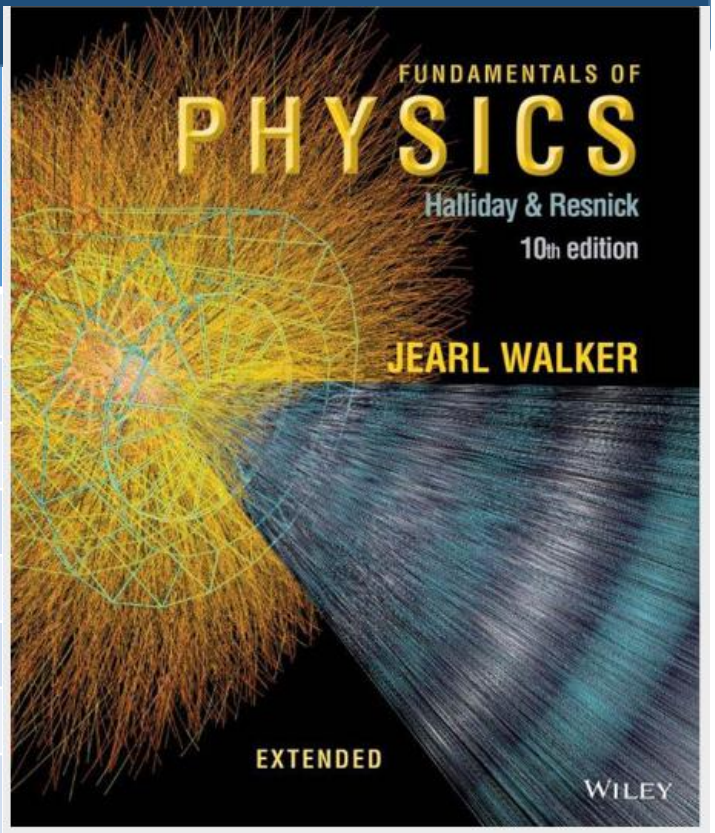
Fundamental Physics II
Electricity – Chapter 1

For B1 Students
Academic year 2024-2025

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OVERVIEW

Chapter	Contents	Hours			Ref./Resources
		Lect.	Exr.	Prc.	
1	Coulomb's law*	2.0	1.0		
2	Electric field*	2.0	1.0		
3	Gauss' law	2.0	1.0		
4	Electric potential	2.0	1.0		
5	Magnetic force and magnetic field	2.0	1.0		
6	Biot-Savart's law and Ampere's law	2.0	1.0		
7	Induction	2.0	1.0		
8	Motion of charge in electric and magnetic fields	2.0	1.0		
		16	8		



Textbook:

[1] J. Parker, D. Halliday, R. Resnick, Fundamentals of physics 10th Edition, Wiley, IBSN 978-1-118-23072-5 (2007).

Assessment/ Evaluation		
	Attendance/Attitude	**10%**
	Exercise(s)	0%
	Practicals	0%
	Mid-term test	**30 %**
	Final exam	**60 %**

Chapter 1. Coulomb's law (2.0h)

1.1 Coulomb's law

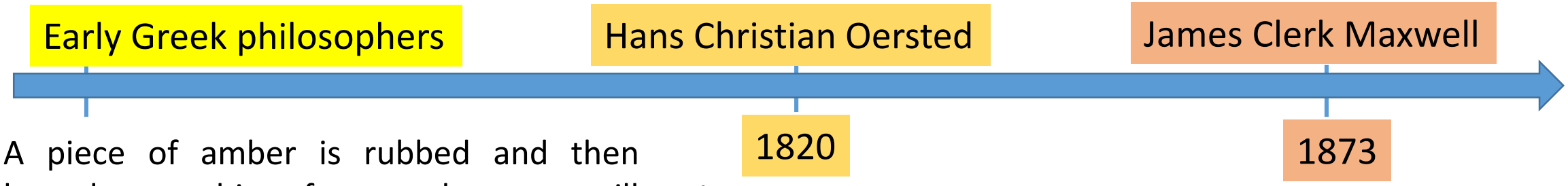
1.2 Charge is quantized

1.3 Charge is conserved

List of Exercises:

In textbook "J. Parker, D. Halliday, R. Resnick, Fundamentals of physics 10th Edition, Wiley, IBSN 978-1-118-23072-5 (2007)."

21. 9, 10, 15, 22, 26, 75



- ✓ A piece of amber is rubbed and then brought near bits of straw, the straw will jump to the amber
- ✓ A certain type of stone is brought near bits of iron, the iron will jump to the stone.

Electricity – magnetism:
An electric current in a wire can deflect a magnetic compass needle.

From Faraday's ideas into mathematical form.

Chapter 1. Coulomb's law (2.0h)

1.1 Coulomb's law

1.1.1 Electric charge

Two types of charge
(named by Benjamin Franklin)

positive ⊕ negative ⊖

- ✓ In most everyday objects, there are about equal numbers of negatively charged particles and positively charged particles: the net charge is zero

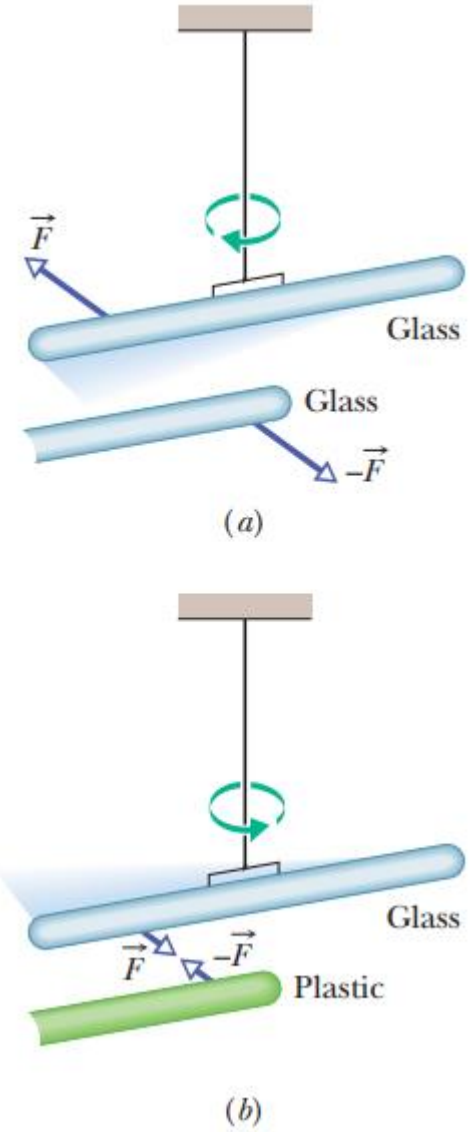
⇒ The charge is said to be ***balanced***

⇒ The object is said to be ***electrically neutral*** (***neutral***)

Excess charge:

- ✓ Objects ***gain*** negative charge and become ***negatively*** charged.
- ✓ Objects ***lose*** negative charge and become ***positively*** charged.

=> The extra charge is said to be an ***excess charge***.



(a) The two glass rods were each rubbed with a silk cloth and one was suspended by thread. When they are close to each other, they repel each other.

(b) The plastic rod was rubbed with fur. When brought close to the glass rod, the rods attract other.

Chapter 1. Coulomb's law (2.0h)

1.1.2 Conductors and Insulators

Conductors

Materials through which charge can move rather freely.

Exp.: Metals, human body, tap water…

Nonconductors (insulators)

Materials through which charge cannot move freely.

Exp.: Rubber, plastic, glass, pure water...

Semiconductors

Materials are intermediate between conductors and insulators.

Exp.: silicon, germanium…

Superconductors

Materials are perfect conductors, allowing charge to move without any hindrance.

Conducting path

Charge is transferred between objects.

A pathway of conductors between an object and Earth's surface: ***ground*** the object

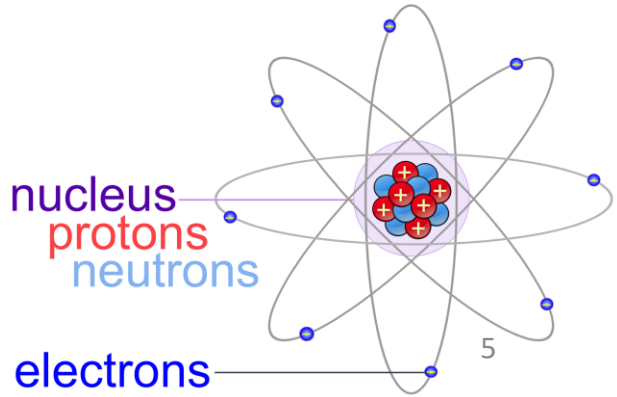
Neutralizing the object (by eliminating an unbalanced positive and negative charge): ***discharge*** the object.

Charged particles

Atoms: positively charged protons
negatively charged electrons
electrically neutral neutrons

Positive ions: atoms lose electrons. Mobile electrons are called conduction electrons.

Negative ions: atoms gain electrons



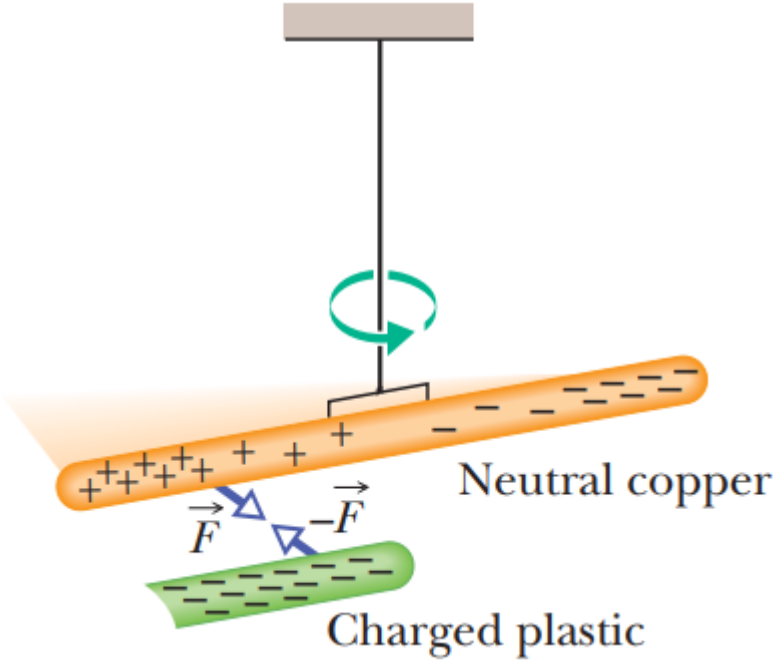
Chapter 1. Coulomb's law (2.0h)

1.1.2 Conductors and Insulators

Induced charge

The neutral copper rod (see Figure) is said to have an ***induced charge***, i.e., some of its positive and negative charges have been separated due to the presence of a nearby charge.

? What happens if a **positively** charged glass rod is brought near one end of a neutral copper rod?



A neutral copper rod is electrically isolated from its surroundings by being suspended on a nonconducting thread. Either end of the copper rod will be ***attracted*** by a charged rod.

Here, conduction electrons in the copper rod are ***repelled*** to the far end of that rod by the negative charge on the plastic rod.

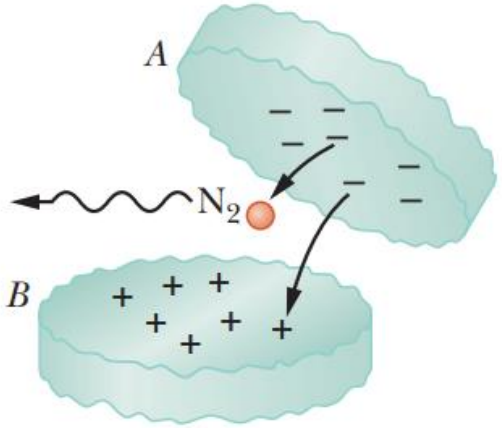
Then that negative charge attracts the remaining positive charge on the near end of the copper rod, rotating the copper rod to bring that near end closer to the plastic rod.

Blue flashes from a wintergreen lifesaver

A crystal breaks into pieces *A* and *B*, with *A* ending up with more electrons on its surface than *B*.

=> *B* has positive ions.

=> Electrons on *A* are strongly attracted to the positive ions on *B*



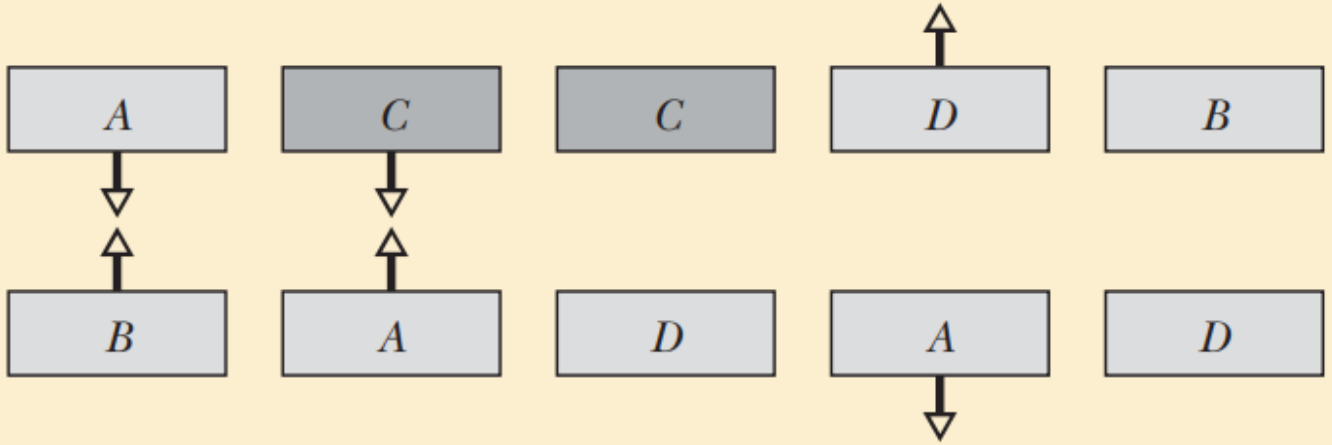
Two pieces of a wintergreen LifeSaver candy as they fall away from each other. Electrons jumping from the negative surface of piece *A* to the positive surface of piece *B* collide with nitrogen molecules in the air.

=> Emit ultraviolet light

=> The wintergreen molecules on the surfaces of the candy pieces absorb the UV light and then emit blue light!

Chapter 1. Coulomb's law (2.0h)

Checkpoint 1.1: Conductors and Insulators
The figure shows five pairs of plates: *A*, *B*, and *D* are charged plastic plates and *C* is an electrically neutral copper plate. The electrostatic forces between the pairs of plates are shown for three of the pairs. For the remaining two pairs, do the plates repel or attract each other?



- *A* and *B*: attract

⇒ Differently charged plates
(for exp.: *A*: positive; *B*: negative)

- *A* and *D*: repel

=> Same charged plates (*A*: positive; *D*: positive)

=> *B* and *D*: attract

=> *D* and *C*: attract

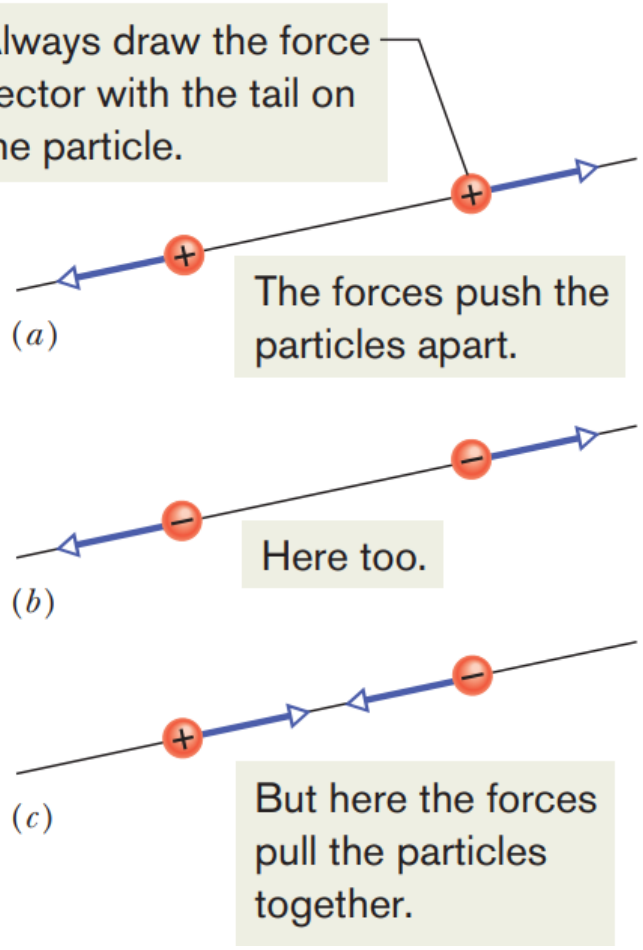
Chapter 1. Coulomb's law (2

1.1.3 Coulomb's law

- ✓ The equation for the electrostatic force acting on the particles: ***Coulomb's law***.
- ✓ Works for only charged particles.
- ✓ Two charged particles are brought near each other, they each exert an **electrostatic force** on the other.
  - If the particles have the same sign of charge: **repel** each other.

  => The force vector on each is directly *away* from the other particle.
  - If the particles have the opposite sign of charge: **attract** each other.

  => The force vector on each is directly *toward* the other particle.



Two charged particles repel each other if they have the same sign of charge, either (a) both positive or (b) both negative.
(c) They attract each other if they have opposite signs of charge.

Chapter 1. Coulomb's law (2.0h)

1.1.3 Coulomb's law

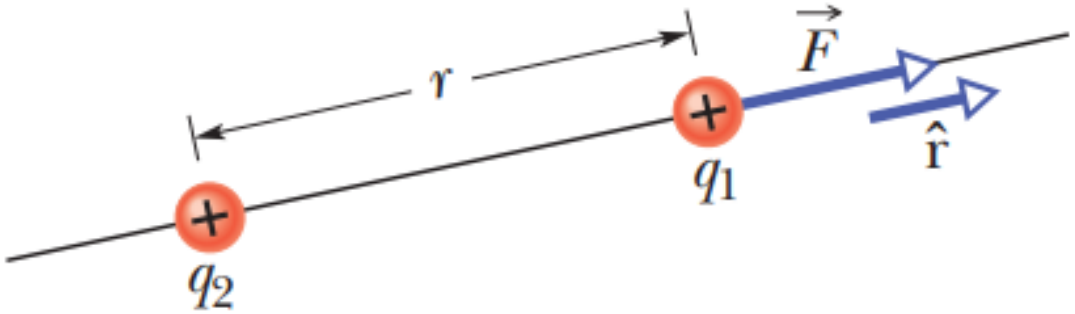
- ✓ Particle 1 has charge $q_1(\text{Coulomb, C})$
- ✓ Particle 2 has charge $q_2(\text{Coulomb, C})$
- ✓ Force acting on particle 1 by particle 2 at a distance $r(\text{m})$:

$$\vec{F}_{12} = k\frac{q_1 q_2}{r^2}\hat{r} \qquad (1.1)$$

- ✓ k: the electrostatic constant – Coulomb constant
- ✓ $\vec{F}(\text{N})$ points radially away from particle 2
- ✓ $\hat{r}$: unit vector, has a magnitude of exactly 1 and no unit
- ✓ $i(\text{Ampere}, A)$: electric current, is the rate at which charge moves past a point or through a region:

$$i = \frac{dq}{dt} \qquad (1.2)$$

$$1\text{C} = (1A)(1s)$$



The electrostatic force on particle 1 can be described in terms of a unit vector $\hat{r}$ along an axis through the two particles, radically away from particle 2.

Force magnitude

$$F_{12} = \frac{1}{4\pi\varepsilon_0}\frac{|q_1||q_2|}{r^2} \qquad (1.3)$$

$$k = \frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^9 \text{N} \cdot \text{m}^2/\text{C}^2 \qquad (1.4)$$

$\varepsilon_0 = 8.85 \times 10^{-12}\text{C}^2/\text{N} \cdot \text{m}^2$: permittivity constant (1.5)

Chapter 1. Coulomb's law (2.0h)

1.1.3 Coulomb's law

Multiple forces

✓ The principle of superposition:

n charged particles near a chosen particle called particle 1; then the net force on particle 1 is given by the vector sum:

$$\vec{F}_{1,\text{net}} = \vec{F}_{12} + \vec{F}_{13} + \vec{F}_{14} + \cdots + \vec{F}_{1n} \quad (1.6)$$

$\vec{F}_{1n}$: the force on particle 1 due to the presence of particle *n*.

Shell theories

✓ **Shell theory 1.** A charged particle outside a shell with charge uniformly distributed on its surface is attracted or repelled as if the shell's charge were concentrated as a particle at its center.

✓ **Shell theory 2.** A charged particle inside a shell with charge uniformly distributed on its surface has no net force acting on it due to the shell.

Spherical conductors

✓ If excess charge is placed on a spherical shell that is made of conducting material, the excess charge spreads uniformly over the (external) surface.

✓ If we remove negative charge from a spherical metal shell, the resulting positive charge of the shell is also spread uniformly over the surface of the shell.

Chapter 1. Coulomb's law (2.0h)

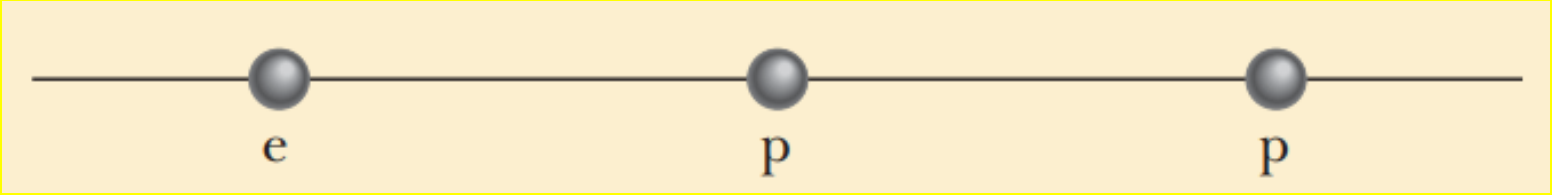
Checkpoint 1.2: Coulomb's law

The figure shows two protons symbol (p) and one electron (symbol e) on an axis. On the central proton, what is the direction of

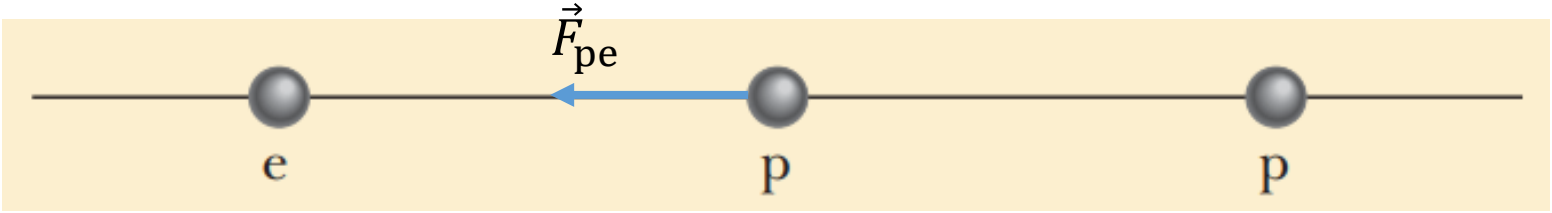
(a) The force due to the electron

(b) The force due to the other proton

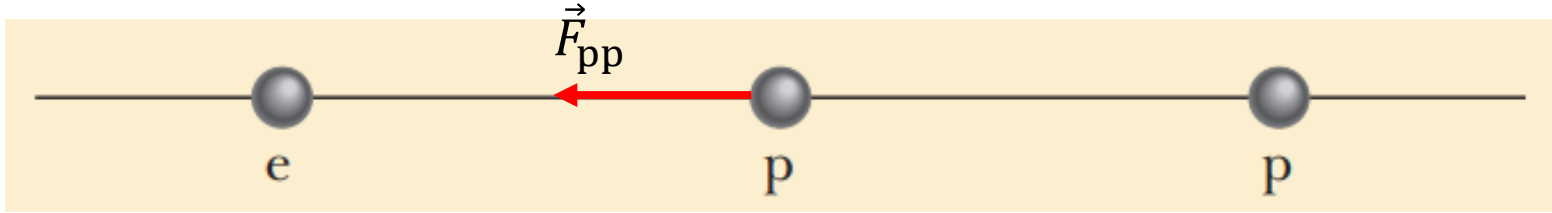
(c) The net force



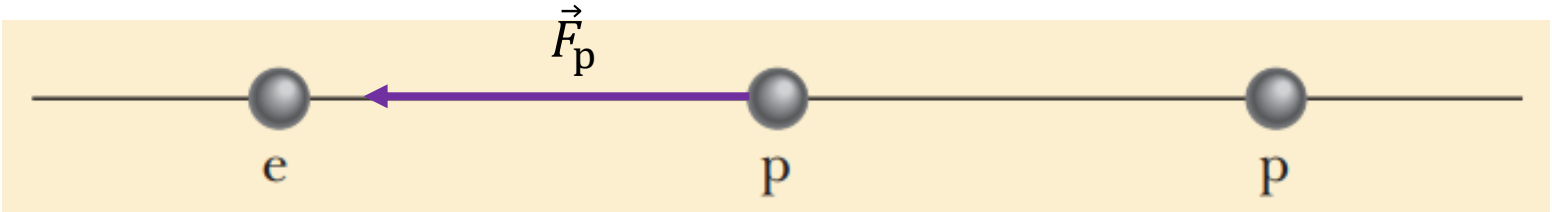
(a) The force due to the electron



(b) The force due to the other proton



(c) The net force



Apply:

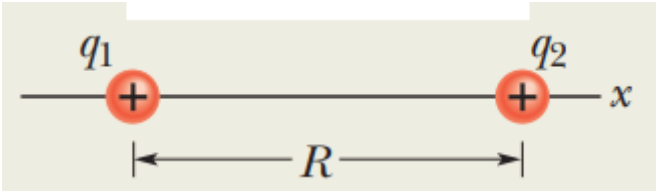
$$\vec{F}_{1,\text{net}} = \vec{F}_{12} + \vec{F}_{13} + \vec{F}_{14} + \cdots + \vec{F}_{1n} \qquad (1.6)$$

$$\vec{F}_{p} = \vec{F}_{pe} + \vec{F}_{pp}$$

Chapter 1. Coulomb's law (2.0h)

Exp. 1.1: Coulomb's law

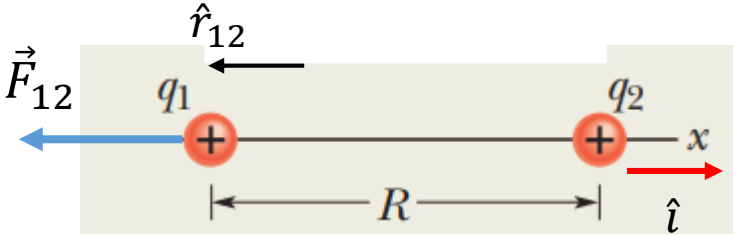
Figure below shows two positively charged particles fixed in place on an *x* axis. The charges are $q_1 = 1.60 \times 10^{-19}\text{C}$, $q_2 = 3.20 \times 10^{-19}\text{C}$, and the particle separation is $R = 0.0200\ \text{m}$. What are the magnitude and direction of the electrostatic force $\vec{F}_{12}$ on particle 1 from particle 2?



The electrostatic force $\vec{F}_{12}$ on particle 1 from particle 2:

$\vec{F}_{12} = k\dfrac{q_1 q_2}{r^2}\hat{r}_{12}$ is directly *away* from particle 2, in the negative direction of the *x* axis.

$$F_{12} = \frac{1}{4\pi\varepsilon_0}\frac{|q_1||q_2|}{r^2} = k\frac{|q_1||q_2|}{r^2} = 8.99 \times 10^9 \frac{1.60 \times 10^{-19} \times 3.20 \times 10^{-19}}{0.02^2} = 1.15 \times 10^{-24}(\text{N})$$



$$\vec{F}_{12} = (-1.15 \times 10^{-24}\text{N})\hat{\imath}$$

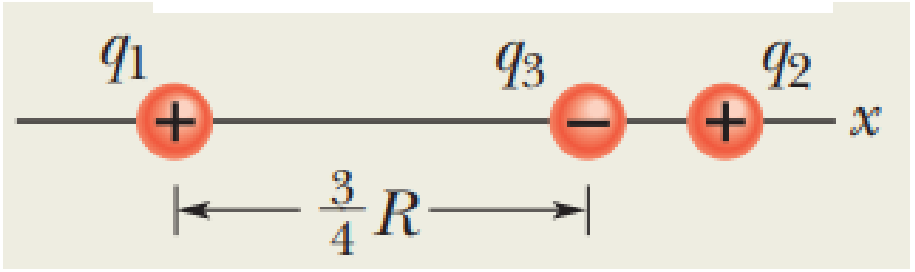
The electrostatic force $\vec{F}_{12}$ on particle 1 from particle 2 has magnitude of $1.15 \times 10^{-24}(\text{N})$ and the direction of 180° (relative to the positive direction of the *x* axis).

Chapter 1. Coulomb's law (2.0h)

Exp. 1.2: Coulomb's law

From the system in Exp. 1.1, adding particle 3. Three charged particles fixed in place on an *x* axis. The charges are $q_1 = 1.60 \times 10^{-19}\text{C}$, $q_2 = 3.20 \times 10^{-19}\text{C}$, $q_3 = -3.20 \times 10^{-19}\text{C}$ and particle 3 is at a distance $\frac{3}{4}R$ from particle

1. What is the net electrostatic force $\vec{F}_{1,\text{net}}$ on particle 1 due to particle 2 and 3?



The electrostatic force $\vec{F}_{12}$ on particle 1 from particle 2:

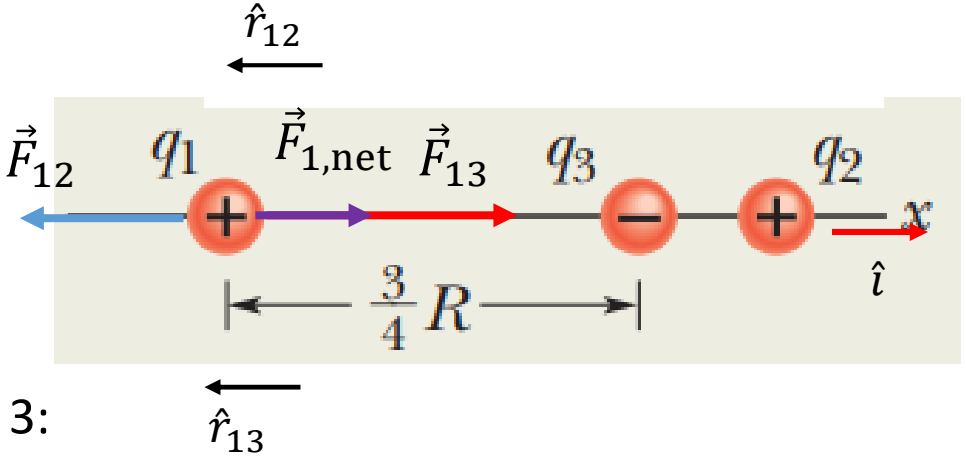
$$\vec{F}_{12} = k\frac{q_1 q_2}{r_{12}^2}\hat{r}_{12} = -k\frac{q_1 q_2}{r_{12}^2}\hat{\imath} = -1.15 \times 10^{-24}\text{N}\hat{\imath}$$

The electrostatic force $\vec{F}_{13}$ on particle 1 from particle 3:

$$\vec{F}_{13} = k\frac{q_1 q_3}{r_{13}^2}\hat{r}_{13} = -k\frac{q_1 q_3}{r^2}\hat{\imath}$$

$$= -8.99 \times 10^9 \frac{1.60 \times 10^{-19} \times (-3.20 \times 10^{-19})}{0.015^2}\hat{\imath}$$

$$= (2.05 \times 10^{-24}\text{N})\hat{\imath}$$



The net electrostatic force $\vec{F}_{1,\text{net}}$ on particle 1 due to particle 2 and 3:

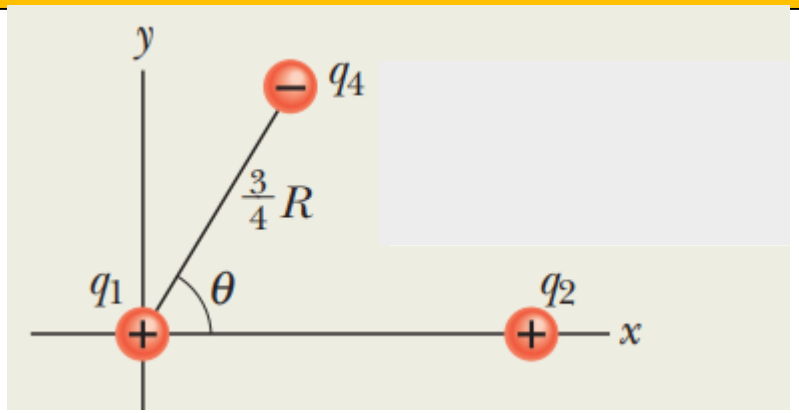
$$\vec{F}_{1,\text{net}} = \vec{F}_{12} + \vec{F}_{13}$$
$$= (-1.15 \times 10^{-24}\text{N})\hat{\imath} + (2.05 \times 10^{-24}\text{N})\hat{\imath}$$
$$= (9 \times 10^{-25}\text{N})\hat{\imath}$$

The magnitude of $9 \times 10^{-25}(\text{N})$ the direction of 0° (relative to the positive direction of the *x* axis).

Chapter 1. Coulomb's law (2.0h)

Exp. 1.3: Coulomb's law

From the system in Exp. 1.1, adding particle 4. The charges are $q_1 = 1.60 \times 10^{-19}\text{C}$, $q_2 = 3.20 \times 10^{-19}\text{C}$, $q_4 = -3.20 \times 10^{-19}\text{C}$; particle 4 is at a distance $\frac{3}{4}R$ from particle 1 and lies on a line that makes an angle $\theta = 60°$ with the *x* axis. What is the net electrostatic force $\vec{F}_{1,\text{net}}$ on particle 1 due to particle 2 and 4?

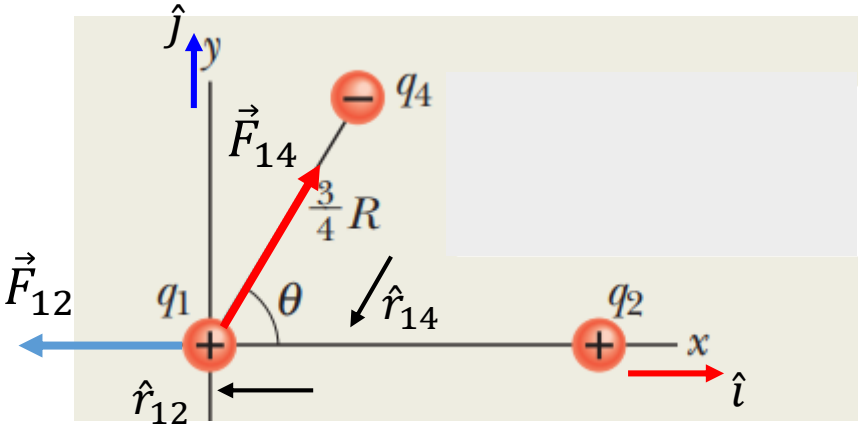


The electrostatic force $\vec{F}_{12}$ on particle 1 from particle 2:

$$\vec{F}_{12} = k\frac{q_1 q_2}{r_{12}^2}\hat{r}_{12} = -k\frac{q_1 q_2}{r_{12}^2}\hat{\imath} = (-1.15 \times 10^{-24}\text{N})\hat{\imath}$$

The electrostatic force $\vec{F}_{14}$ on particle 1 from particle 4:

$$\vec{F}_{14} = k\frac{q_1 q_4}{r_{14}^2}\hat{r}_{14} = -k\frac{q_1 q_4}{r_{14}^2}\cos 60°\hat{\imath} - k\frac{q_1 q_4}{r_{14}^2}\sin 60°\hat{\jmath}$$
$$= 2.05 \times 10^{-24}\text{N}(0.5\hat{\imath} + 0.866\hat{\jmath})$$
$$= (1.025 \times 10^{-24}\text{N})\hat{\imath} + (1.775 \times 10^{-24}\text{N})\hat{\jmath}$$



The net electrostatic force $\vec{F}_{1,\text{net}}$ on particle 1 due to particle 2 and 4:

$$\vec{F}_{1,\text{net}} = \vec{F}_{12} + \vec{F}_{14}$$
$$= (-1.15 \times 10^{-24}\text{N})\hat{\imath} + (1.025 \times 10^{-24}\text{N})\hat{\imath} + (1.775 \times 10^{-24}\text{N})\hat{\jmath}$$
$$= (-0.125 \times 10^{-24}\text{N})\hat{\imath} + (1.775 \times 10^{-24}\text{N})\hat{\jmath}$$

Chapter 1. Coulomb's law (2.0h)

Checkpoint 1.3: Coulomb's law

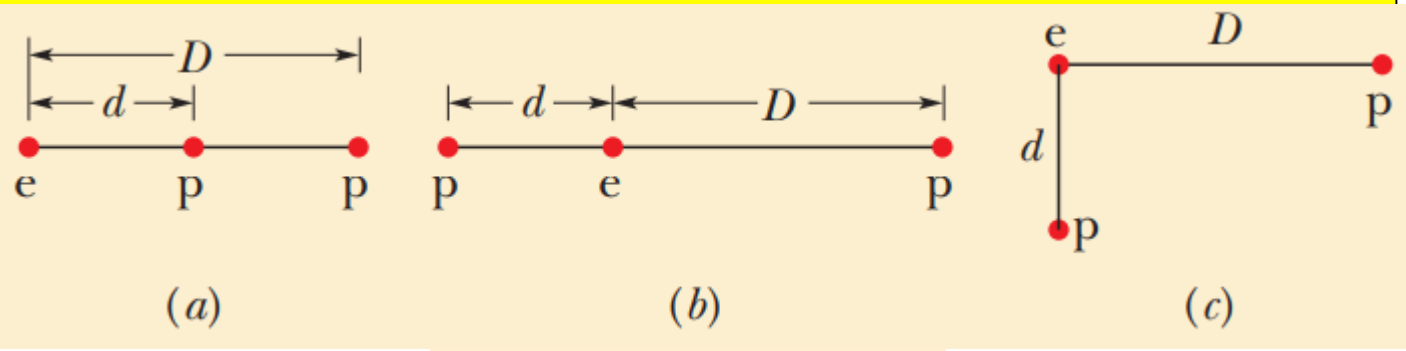
The figure here shows three arrangements of an electron e and two protons.

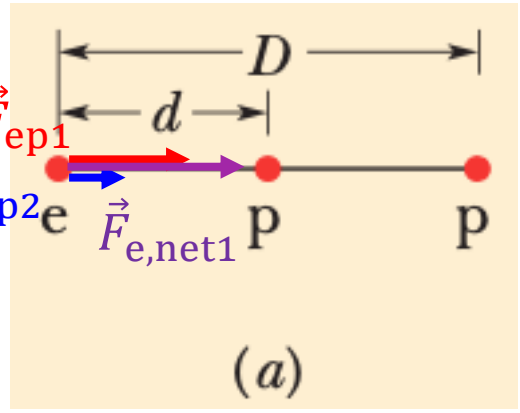
(a) Rank the arrangements according to the magnitude of the net electrostatic force on the electron due to the protons, largest first.

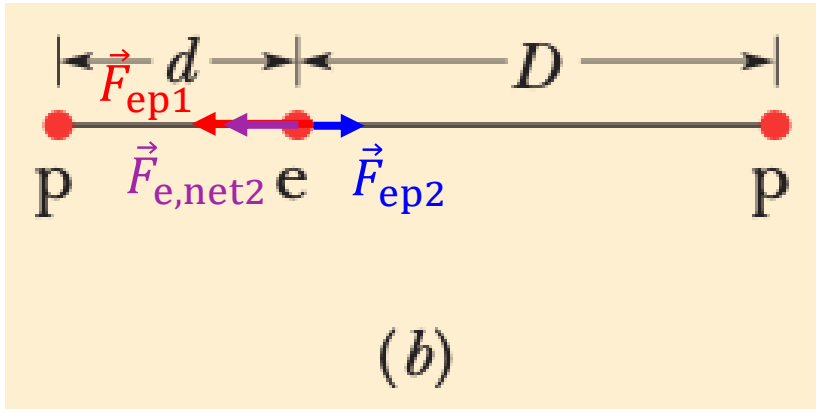
(b) In situation *c*, is the angle between the net force on the electron and the line labeled *d* less than or more than 45°.

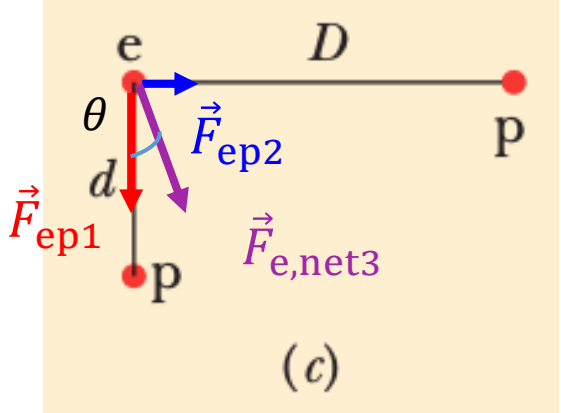
$\vec{F}_{e,net} = \vec{F}_{ep1} + \vec{F}_{ep2} \quad D = 2d \Rightarrow |F_{ep1}| = 4|F_{ep2}|$

$F_{e,net}^2 = F_{ep1}^2 + F_{ep2}^2 + 2F_{ep1}F_{ep2}\cos(\vec{F}_{ep1}, \vec{F}_{ep2})$









$\tan\theta = \dfrac{F_{ep2}}{F_{ep1}} = \dfrac{1}{4}$

$\theta = 14° < 45°$

$F_{e,net1}^2 = F_{ep1}^2 + F_{ep2}^2 + 2F_{ep1}F_{ep2}\cos 0° \quad F_{e,net2}^2 = F_{ep1}^2 + F_{ep2}^2 + 2F_{ep1}F_{ep2}\cos 180° \quad F_{e,net3}^2 = F_{ep1}^2 + F_{ep2}^2 + 2F_{ep1}F_{ep2}\cos 90°$

$F_{e,net1} = F_{ep1} + F_{ep2} \qquad F_{e,net2} = F_{ep1} - F_{ep2} \qquad F_{e,net3} = \sqrt{F_{ep1}^2 + F_{ep2}^2}$

Chapter 1. Coulomb's law (2.0h)

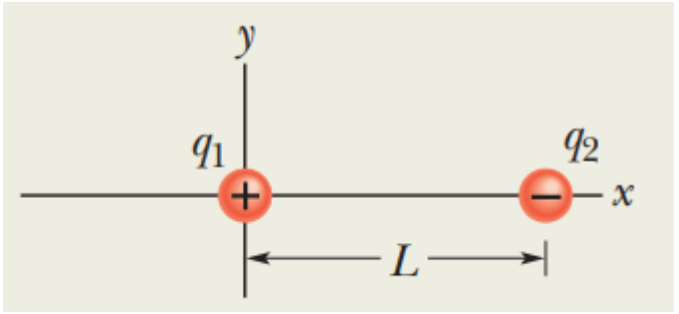
Exp. 1.4: Equilibrium of two forces on a particle

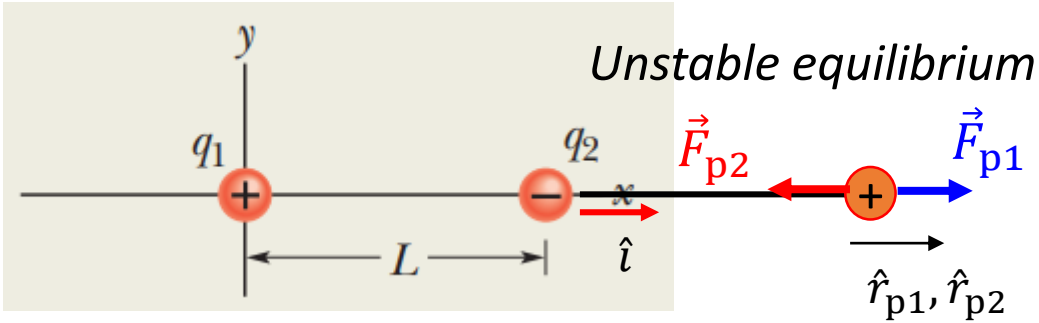
Figure below shows two particles fixed in place: a particle of charge $q_1 = +8q$ at the origin and a particle of charge $q_2 = -2q$ at $x = L$. At what point (other than infinitely far away) can a proton be placed so that it is in *equilibrium* (the net force on it is zero)? Is that equilibrium *stable* or *unstable*? (That is, if the proton is displaced, do the forces drive it back to the point of equilibrium or drive it farther away?)

$$\vec{F}_{\text{p,net}} = \vec{F}_{\text{p1}} + \vec{F}_{\text{p2}} \qquad e = 1.602 \times 10^{-19}\text{C}$$

$$\vec{F}_{\text{p1}} = k\frac{q_1 p}{r_{\text{p1}}^2}\hat{r}_{\text{p1}} = k\frac{8qe}{r_{\text{p1}}^2}\hat{r}_{\text{p1}}$$ is directly *away* from particle q_1

$$\vec{F}_{\text{p2}} = k\frac{q_2 p}{r_{\text{p2}}^2}\hat{r}_{\text{p2}} = -k\frac{2qe}{r_{\text{p2}}^2}\hat{r}_{\text{p2}}$$ is directly *toward* particle q_2





Unstable equilibrium

The proton is placed so that it is in *equilibrium*:

$$\vec{F}_{\text{p,net}} = \vec{F}_{\text{p1}} + \vec{F}_{\text{p2}} = \vec{0} \qquad k\frac{8qe}{r_{\text{p1}}^2}\hat{r}_{\text{p1}} = k\frac{2qe}{r_{\text{p2}}^2}\hat{r}_{\text{p2}} \qquad \hat{r}_{\text{p1}} = \hat{r}_{\text{p2}}$$

$$r_{\text{p1}} = 2r_{\text{p2}} \qquad r_{\text{p1}} = L + r_{\text{p2}} = 2r_{\text{p2}}$$

$$r_{\text{p1}} = 2L; r_{\text{p2}} = L$$

$$\vec{F}_{\text{p1}} = -\vec{F}_{\text{p2}} \qquad \frac{4}{r_{\text{p1}}^2}\hat{r}_{\text{p1}} = \frac{1}{r_{\text{p2}}^2}\hat{r}_{\text{p2}} \qquad \frac{4}{r_{\text{p1}}^2} = \frac{1}{r_{\text{p2}}^2}$$

=> The proton is placed outside the line connecting q_1 and q_2

Chapter 1. Coulomb's law (2.0h)

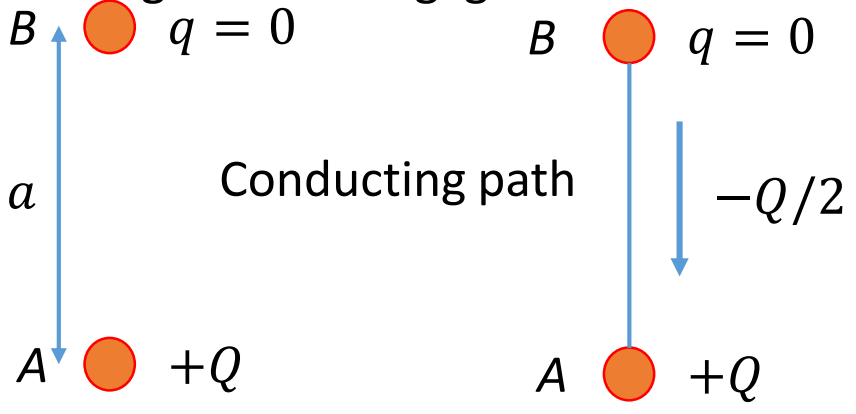
Exp. 1.5: Charge sharing by two identical conducting spheres

Figure below shows two identical, electrically isolated conducting spheres *A* and *B* are separately by a (center-to-center) distance *a* that is large compared to the spheres. Sphere *A* has a positive charge of +*Q*, and sphere *B* is electrically neutral. Initially, there is no electrostatic force between the spheres. (The large separation means there is no induced charge.)

(a) Suppose the spheres are connected for a moment by a conducting wire. The wire is thin enough so that any net charge on it is negligible. What is the electrostatic force between the spheres after the wire is removed?

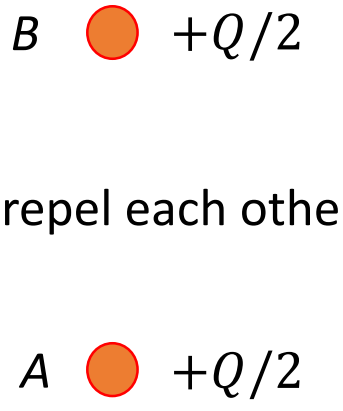
(b) Next, suppose sphere *A* is grounded momentarily, and then the ground connection is removed. What now is the electrostatic force between the spheres?

(a) Suppose the spheres are connected for a moment by a conducting wire. The wire is thin enough so that any net charge on it is negligible. The electrostatic force between the spheres after the wire is removed is



Identical, electrically isolated conducting spheres *A* and *B*

A and *B* are connected:
Electrons move from *B* to *A* until identical charges: $+Q/2$



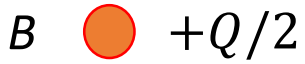
The first shell theory:

$$F = \frac{1}{4\pi\varepsilon_0}\frac{|q_A||q_B|}{r^2}$$

$$F = \frac{1}{4\pi\varepsilon_0}\frac{\frac{Q}{2}\frac{Q}{2}}{a^2}$$

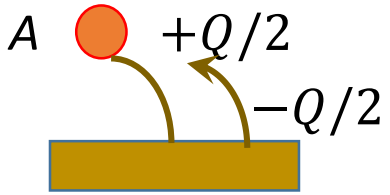
$$= \frac{1}{16\pi\varepsilon_0}\frac{Q^2}{a^2}$$

(b) Next, suppose sphere *A* is grounded momentarily, and then the ground connection is removed. The electrostatic force between the spheres is:





No electrostatic force between the spheres





A is grounded momentarily:
Electrons move from the ground to *A* until *A* is neutralized.

Chapter 1. Coulomb's law (2.0h)

1.2 Charge is quantized

✓ Any positive or negative charge q that can be detected can be written as

$$q = ne \qquad n = \pm 1, \pm 2, \pm 3, \ldots, \qquad (1.7)$$

in which e is the **elementary charge**

$$e = 1.602 \times 10^{-19}\text{C} \qquad (1.8)$$

Particle	Symbol	Charge
Electron	e or e^-	$-e$
Proton	p	$+e$
Neutron	n	0

✓ When charge can have only discrete values rather than any value: is **quantized**

Checkpoint 1.4: Charge is quantized
Initially, sphere *A* has a charge of -50*e* and sphere *B* has a charge of +20*e*. The spheres are made of conducting material and are identical in size. If the spheres then touch, what is the resulting charge on sphere *A*?

B $q_B = +20e$

The spheres are made of conducting material and are identical in size.

A $q_A = -50e$

B

A

The spheres touch…

Electrons move from *B* to *A* until $q'_A = q'_B = \frac{q_A + q_B}{2} = 15e$

Chapter 1. Coulomb's law (2.0h)

Exp. 1.5: Mutual electric repulsion in a nucleus

The nucleus in an iron atom has a radius of about $4.0 \times 10^{-15}\text{m}$ and contains 26 protons.

(a) What is the magnitude of the repulsive electrostatic force between two of the protons that are separated by $4.0 \times 10^{-15}\text{m}$?

(b) What is the magnitude of the gravitational force between those same two protons?

(a) The magnitude of the repulsive electrostatic force between two of the protons that are separated by $4.0 \times 10^{-15}\text{m}$ is

$$p = +e = 1.602 \times 10^{-19}\text{C}$$

$$F_E = \frac{1}{4\pi\varepsilon_0}\frac{|q_1||q_2|}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{(1.602 \times 10^{-19})^2}{(4.0 \times 10^{-15})^2} = 14(\text{N})$$

Particle	Symbol	Charge
Electron	e or e^-	$-e$
Proton	p	$+e$
Neutron	n	0

$$e = 1.602 \times 10^{-19}\text{C}$$

(b) The magnitude of the gravitational force between those same two protons

$$F_G = G\frac{m_\text{p} m_\text{p}}{r^2} = 6.67 \times 10^{-11}\frac{(1.67 \times 10^{-27})^2}{(4.0 \times 10^{-15})^2} = 1.2 \times 10^{-35}(\text{N})$$

$$F_E \gg F_G$$

Chapter 1. Coulomb's law (2.0h)

1.3 Charge is conserved

Conservation of charge

- ✓ Exp.: Rub a glass rod with silk:

  The glass rod becomes positively charged

  The silk becomes negatively charged

  Same magnitude $|q_{\text{rod}}| = |q_{\text{silk}}|$

- ✓ Radioactive decay:

$$^{238}\text{U} = {}^{234}\text{Th} + {}^{4}\text{He}$$

^{238}U contains 92 protons: $q_{\text{U}} = +92e$

^{234}Th contains 90 protons: $q_{\text{Th}} = +90e$

^{4}He contains 2 protons: $q_{\text{He}} = +2e$

} Charge is conserved

- ✓ Annihilation:

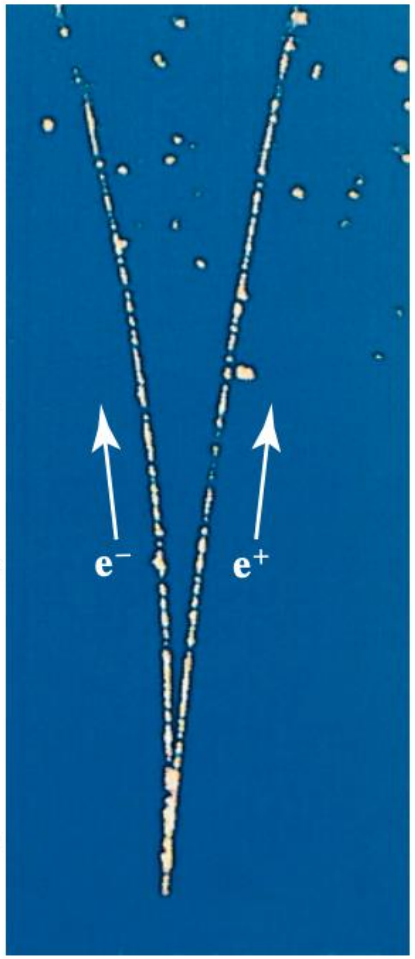
  $e^- + e^+ \rightarrow \gamma + \gamma$

  An electron and a positron transform into two gamma rays.

- ✓ Pair production:

  $\gamma \rightarrow e^- + e^+$

  A gamma ray transforms into an electron and a positron.



Courtesy Lawrence Berkeley Laboratory

A photograph of trails of bubbles left in a bubble chamber by an electron and a positron. The pair of particles was introduced by a gamma ray that entered the chamber directly from the bottom. Being electrically neutral, the gamma ray did not generate a telltale trail of bubbles along its path, as the electron and positron did.

“When you really want something the whole universe conspires in helping you to achieve it”
(The Alchemist, Paulo Coelho)

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GOOD LUCK and THANK YOU

Make it Simple but Significant!